Roadmap. Srednicki §18: power counting, $D = [T_E] - \sum_n V_n[g_n]$, sorts theories by the sign of $[g]$; φ^3 in $d = 6$ needs only Y, A, B, C. §19 (PDF pp. 5–6): skeletons make $2 \to 2$ a finite tree sum of exact $\tilde{\Delta}$, V_3, V_4. §20 (pp. 6–8): T at one loop; its high-energy $\ln(s/m^2)$ opens LecNote15 (§26), and the RG resums it (§§27–28). Supplements: DimensionalAnalysis_hbar, FeynmanRulesLSZ, iPiassignment, RenormalizationScaleGroup, Renormalizability, OneLoopHighEnergy.

Renormalizability.

Now we are armed with loop insights,

The superficial divergences of any loop diagram

$$D = d \cdot L - 2I.$$

(d: dim; L: loops; I: internal propagators.)

If $D \geq 0$, it is divergent.

But this doesn't tell us much about the theory.

Now. ~~Fig. 7.E.~~ for n external legs.

$$[\text{diagram}] = [g_E] = dL - 2I + \sum_{n=3}^{\infty} V_n [g_n].$$

Skip this counting

$$D = [g_E] - \sum_{n=3}^{\infty} V_n [g_n].$$

Next page is better

For whatever $[g_E]$, there will always exist a large enough V_n to make D positive (semi) if $[g_n]$ is negative

$\therefore$ $[g_n]$ negative means the theory is non-renormalizable

A given # of external legs fixes the dimension as

$$iT = \text{leading-order} + \text{Next-leading order} + \text{higher order}$$

(typically)

$$iT = \underline{\text{tree}} + \underline{\text{one-loop}} + \underline{\text{two-loops}} + \cdots$$

units have to be the same.

So for a fixed # of external legs

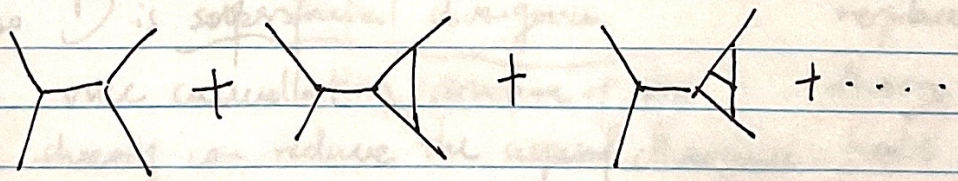
$$[\text{diagrams}] = [\text{external legs}] = dL - 2I + \sum_{\{n\}} V_n [g_n]$$

(~~tree~~ fixed by # of)

Note that n can be any consistent # of vertices.

For the example of three point interaction,

for the 2→2 process, we have



On the other hand $D = dL - 2I$.

So

$D = [\text{external legs}] - \sum_{\{n\}} V_n [g_n]$ (1)

where V_n can keep growing.

Clearly, if $[g_n]$ is negative, there will always exist a large enough V_n to make $D \geq 0$ and hence divergent.

So if any $[g] < 0$, the theory need to be renormalized infinite amount of times

Hence "Non-renormalizable".

↓

but this not that terrible thing

Effective theory

The theory can be replaced by another theory at some high scale. (2)

Also D is superficial divergence.

nice cancellations / structure of the theory can reduce the ~~superficial~~ divergence.

1 [external legs] is the dimension of the E-point amplitude, $[T_E] = d - \frac{E}{2}(d-2) = [g_E]$: a lone φ^E vertex gives $T_E = g_E$. For φ^3 in $d = 6$, $[g] = 0$, so $D = 6 - 2E$ at every order: only $E = 1, 2, 3$ diverge, cancelled by Y; A, B; $C = Z_g - 1$. Each label on PDF p. 4 equals $[T_4] = 4 - d$.

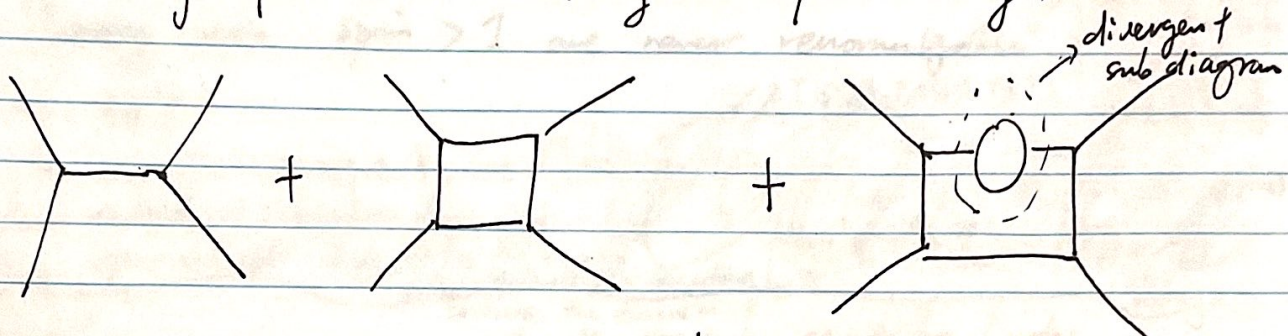
2 Not terrible: a coupling with $[g] < 0$ is $g \sim \Lambda^{[g]}$, Λ the scale of new physics. At momenta $k \ll \Lambda$ each such vertex costs $(k/\Lambda)^{-[g]}$, so to a given accuracy only finitely many couplings and counterterms enter. Fermi theory ($[G_F] = -2$, $\Lambda \sim M_W$) and gravity ($[G_N] = 2 - d$, $\Lambda \sim M_{\rm Pl}$) are such effective theories (§29).

1 Should read $[g] \geq 0$. $[g] > 0$: super-renormalizable, D falls as V grows, so only finitely many diagrams are superficially divergent. $[g] = 0$: renormalizable, D depends on E only. $[g] < 0$: non-renormalizable (PDF p. 3).

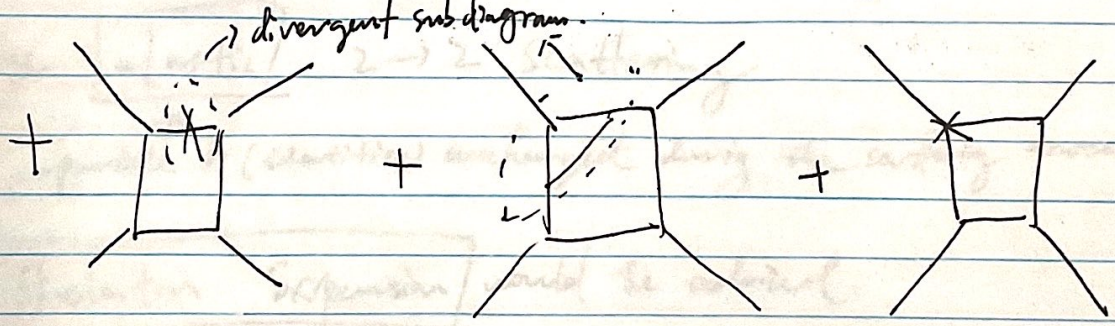
for $[g] \leq 0$ (1) { super-renormalizable
renormalizable.

Let's "see" the meaning.

(e.g. for $2 \to 2$ scattering in φ^3 theory)



$2[g] - 2$ $\quad 4[g] - 4\times 2 + d$. $\quad 6[g] - 7\times 2 + 2d$



but these divergences are exactly removed by

Z_φ, Z_m, Z_g

So we only need finite # of counter terms to renormalize the theory.

1 Why spin matters: power counting assumed propagators falling like $1/k^2$ ($1/k$ for spin $\frac{1}{2}$). A massive vector's, $(g^{\mu\nu} + k^\mu k^\nu/m^2)/(k^2 + m^2)$, tends to a constant: each line adds 2 to D. Coupled to a conserved current (gauge symmetry) the $k^\mu k^\nu$ term drops out. Spin 2 couples to $T^{\mu\nu}$ (two derivatives): gravity, $[G_N] = 2 - d < 0$.

(5)

The theorem on renormalizability is true for spin-0 and spin-1/2, and for the spin-1 field with gauge symmetry.

↑ to be introduced last part of this class.

Theory with spin > 1 are never renormalizable (1) in $d \geq 4$.

This why GR is not part of QFT also why hadron theories have to be replaced by QCD. It is super of elementary theory

One can Dim analyze it, ass. $[1] =$ kinetic $+$ interaction.

$z[\varphi]$ $A, \gamma^\mu, \partial_\mu, \eta[1]$

$[g] \geq (\quad) = (\quad)$ but $[g]$

Some details of interaction forms need. e.g. spinor and vector indices makes the minimal interaction involve derivatives increase the dimension of the interaction terms. see additional notes.

Now let's take a look at the asymptotic behavior.

For the elastic $2 \to 2$ scattering

↑ particle # (identities) unchanged during the scattering process.

A Skeleton Expension would be advised.

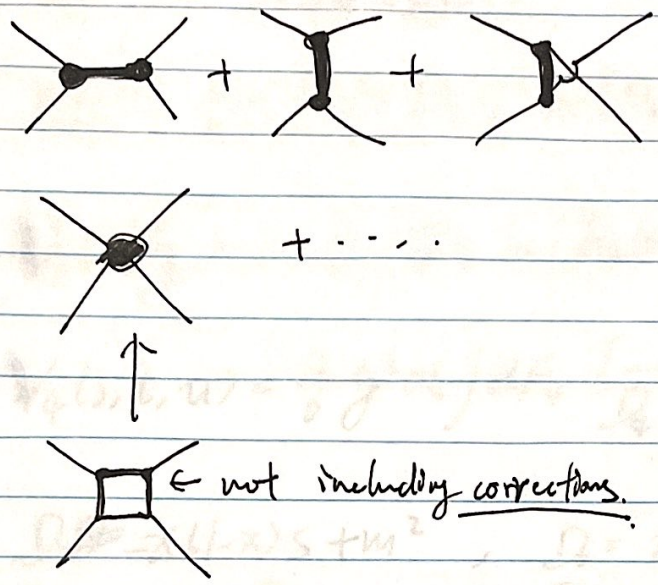
↳ draw all 1PI diagrams that does not contain propagator correction, or vertice correction (for φ^3 theory). 3-point

for subdiagrams.

Then sum all tree-level diagram including the above one for V_n.

(constructed)

Then we use the exact $\tilde{\Delta}(k^2)$ and $\tilde{V}_3(k_1,k_2,k_3)$ such that we are consistent in ordering & counting.



Counting.

tree (g^2)

loop $(g^2 \cdot \frac{g^2}{(4\pi)^3})$ and it is finite. $+\mathcal{O}(g^6)$

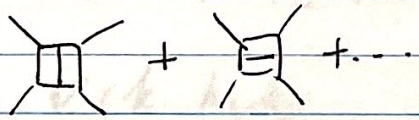
← loop $(g^4 \cdot \frac{1}{(4\pi)^3})$ finite.

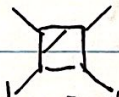
$+\mathcal{O}(g^6)$

← not including corrections.

What about next order?

top-line auto matic

second-line ~~tate~~ includes 

but not 

(as it is already in the first line) (1)

1 Why skeletons: every 1PI subgraph has $D = 6 - 2E$, so only propagator and vertex subgraphs diverge (tadpoles cancel against Y). A skeleton has neither: it is finite, and stays so with the exact, renormalized $\tilde{\Delta}$, V_3 inserted. The graph drawn is 1PI, not an exchange graph: the box with V_3's one-loop part at a corner.

Let's see the amazing result.

$$i\mathcal{T}_{\text{tree}\,(2\to2)} = \frac{1}{i}(ig)^2\left[\tilde{\Delta}(-s)+\tilde{\Delta}(-t)+\tilde{\Delta}(-u)\right].$$

$$i\mathcal{T}_{1\text{-loop}\,(2\to2)} = \frac{1}{i}\Big(\left[i\tilde{V}_3(s)\right]^2\tilde{\Delta}(-s) + \left[i\tilde{V}_3(t)\right]^2\tilde{\Delta}(-t) + \left[i\tilde{V}_3(u)\right]^2\tilde{\Delta}(-u)\Big) + i\tilde{V}_4(s,t,u).$$

You can observe the cyclic property of this amplitude here.

to the given order.

Here we have already calculated every piece:

$$\tilde{\Delta}(-s) = \frac{1}{-s+m^2-\Pi(-s)}$$

1 Should read $\tilde{\Delta}(-s) = 1/(-s + m^2 - \Pi(-s))$ (s is already a mass^2) and $V_3(s)/g = 1 - \frac{1}{2}\alpha \int dF_3 \ln(D_3(s)/m^2)$, with the α of LecNote13 and PDF p. 8.

$$\Pi(s) = \frac{1}{2}\alpha \int_0^1 dx\, D_2(s) \ln[D_2(s)/D_0] - \frac{1}{12}\alpha(-s+m^2).$$

$$V_3(s)/g = 1 - \frac{1}{2}\int dF_3 \ln(D_3(s)/m^2)$$

$$V_4(s,t,u) = \frac{1}{6} g^2 \alpha \int dF_4 \left[\frac{1}{D_4(s,t)} + \frac{1}{D_4(t,u)} + \frac{1}{D_4(u,s)}\right]$$

it is cyclic, not symmetric.

$$D_2(s) = -x(1-x)s + m^2, \quad D_0 = x(1-x)m^2 + m^2$$

2 Should read $D_0 = [1 - x(1-x)]m^2$: D_2 at $s = m^2$ ($k^2 = -m^2$). With $+x(1-x)m^2$, $\Pi(-m^2) = -0.135\,\alpha \neq 0$ and the on-shell condition fails.

$$D_3(s) = -x_1 x_2 s + [1 - (x_1+x_2)x_3]m^2$$

$$D_4(s,t) = -x_1 x_2 s - x_3 x_4 t + [1 - (x_1+x_2)(x_3+x_4)]m^2.$$

↑ symmetric ↑

Here we get $V_3(s)$ from $V(k_1,k_2,k_3)$

by letting $k_1^2 = -m^2$, $k_2^2 = -m^2$

$k_3 = (k_1+k_2)^2 = -s$.

That's the result. One can further evaluate them using various special functions or numerically.

To gain a bit more intuition, we can do the high energy limit at a fixed scattering angle, such that

$$s \gg m^2, \quad |t| \gtrsim m^2, \quad |u| \gtrsim m^2,$$

(recall $s+t+u = 4m^2$)

$$t, u < 0.$$

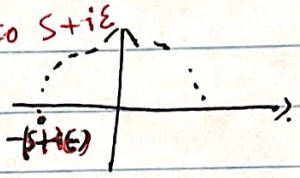
In such limit, we have.

$$\Pi(-s) = -\frac{1}{12}\alpha s\left[\ln(-s/m^2) + 3 - \pi\sqrt{3}\right].$$

$$\Delta(-s) = \frac{1}{-s - \Pi(s)} = -\frac{1}{s}\left(1 + \frac{1}{12}\alpha\left[\ln(-s/m^2) + 3 - \pi\sqrt{3}\right]\right) + O(\alpha^2)$$

↑ recall $+m^2 - i\varepsilon$. so $s + i\varepsilon$

$$\ln(-s) = \ln s - i\pi$$



$$\ln(-t) = \ln|t|, \quad \ln(t) = \ln|t| + i\pi.$$

$$V_3(s)/g = 1 - \frac{1}{2}\alpha\int dF_3\left[\ln(-s/m^2) + \ln(x_1 x_2)\right]$$

$$= 1 - \frac{1}{2}\alpha\left[\ln(-s/m^2) - 3\right]$$

$$\int \frac{dF_4}{D_4(s,t)} = -\frac{3}{s+t}\left(\pi^2 + [\ln(s/t)]^2\right) = \frac{3}{u}\left[\pi^2 + \ln^2(s/t)\right]$$

We get that in the high energy limit that.

$$\mathcal{T}_{1\text{-loop}} = g^2\left[F(s,t,u) + F(t,u,s) + F(u,s,t)\right]$$

↑ cyclic property.

$$F(s,t,u) = -\frac{1}{s}\left(1 - \frac{11}{12}\alpha\left[\ln(-s/m^2) + c\right] - \frac{1}{2}\alpha\ln^2(s/u)\right)$$

$c = (6\pi^2 + \pi\sqrt{3} - 39) = \boxed{2.33}$ (1)

$-\frac{11}{12}\alpha \times 2.33 \simeq -2\alpha$

If α is large (2)

this correction term is bigger than the leading tree level result. (a warning sign of perturbative interpretation)

tree

1 Should read $c = (6\pi^2 + \pi\sqrt{3} - 39)/11 = 2.33$; without the /11 the left side is 25.7. c collects the constants: $-\frac{11}{12}c = 3 + \frac{3-\pi\sqrt{3}}{12} - \frac{\pi^2}{2}$, from V_3^2, $\tilde{\Delta}$ and the box.

2 The warning is the logarithm, not α: $\frac{11}{12}\alpha\ln(s/m^2) = 1$ at $\sqrt{s} = m\,e^{6/(11\alpha)}$, 234 m for $\alpha = 0.1$ ($\frac{11}{12} = 1 - \frac{1}{12}$: V_3^2 and $\tilde{\Delta}$). The m^2 comes from the conditions fixing A, B, C (on shell; $V_3(0,0,0) = g$). LecNote15 (§§27–28) trades it for μ: the observable carries $\ln(s/\mu^2)$, small for $\mu \sim \sqrt{s}$, and $\alpha(\mu)$ runs.